

# FEG Prism

Kuratowski Calculus Engine — Code Manual

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Juan Pablo Silva Alvarado

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*Part of Modulo Synthesis / Fractal Entropic Geometrodynamics*

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*“The universe is strictly computable, finite, and structurally  
homeostatic. The rest is just counting.”*



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# Theory

## Chapter Overview

- *The physical framework: Fractal Entropic Geometrodynamics (FEG)*
- *Three Geometries of spacetime, matter, and logic*
- *Key invariants enforced by the simulation*

## 1.1 Fractal Entropic Geometrodynamics

FEG Prism is a Monte Carlo engine implementing the Kuratowski Calculus — the mathematical framework underlying *Fractal Entropic Geometrodynamics* (FEG) by Juan Pablo Silva Alvarado (DOI: 10.5281/zenodo.18769707). The central claim of FEG is that four-dimensional spacetime, the Standard Model particle spectrum, and quantum mechanics all emerge from a single discrete structure: the Hasse diagram of a Poisson-sprinkled causal set, constrained only by **planarity preservation** (Kuratowski’s theorem).

The engine is named after its fundamental topological particle: the **Causal Prism**  $K_{2,N}$ , a complete bipartite subgraph with two poles and  $N \geq 3$  intermediaries. These are the only non-trivial substructures permitted by the triangle-free property of the transitive reduction.

## 1.2 The Three Geometries

### Definition 1: Geometry of Order — Phase 1

Poisson sprinkling into a 4D causal diamond produces  $N$  events. The **transitive reduction** (Hasse diagram) removes all redundant causal links: edge  $(u, v)$  survives iff no intermediate  $w$  satisfies  $u \prec w \prec v$ . The result is a triangle-free directed acyclic graph (DAG) whose Lorentz invariance is emergent from the isotropy of the Poisson process.

### Definition 2: Geometry of Matter — Phase 2

The **Causal Prism**  $K_{2,N}$  is the unique non-trivial bipartite subgraph permitted in a triangle-free graph. It consists of:

- Two **poles** (origin  $u$ , destination  $v$ ) at graph-distance  $\geq 2$
- $N \geq 3$  **intermediaries**  $w_1, \dots, w_N$  shared by both poles
- Intermediaries are mutually disconnected (guaranteed by triangle-freeness)

Topological mass is the integer  $M = N$  (number of intermediaries). The mass hierarchy reduces to  $N \in \{3, 4, 5+\}$  for electron, muon, tau.

### Definition 3: Geometry of Logic — Computational Complexity

The 2-hop detection algorithm for Causal Prisms runs in  $O(N)$  because the physics is local. Hasse links have bounded proper time ( $\tau \leq 8$ ); therefore  $K_{2,N}$  poles must be within bounded causal distance. The bounded degree ( $\text{MAX\_HASSE\_DEGREE} = 15$ ) ensures  $O(1)$  per-node work. This is not merely an optimisation — it is a computational proof that **physical locality is an information principle**.

### 1.3 Key Invariants

The following invariants are preserved throughout the simulation pipeline:

1. **Triangle-freeness:** The Hasse diagram contains no 3-cliques. Any triangle  $u \rightarrow w \rightarrow v$  with  $u \rightarrow v$  would be transitively reduced. This is a mathematical theorem, not an empirical check.
2. **Planarity preservation:**  $K_5$  threats are absorbed into the nearest pole via vertex contraction, enforcing Kuratowski's planarity criterion. This is the discrete analogue of colour confinement.
3. **Lorentz invariance:** No spatial cutoff is applied during Hasse construction. The light cone search at time offset  $\Delta t$  uses radius  $\Delta t + 1$  (natural units  $c = 1$ ), preserving exact causal structure.
4. **Strict finitism:** All accumulations use u64 integers. The single f64 division occurs only at the final return. No floating-point drift across ensemble realisations.
5. **Deterministic reproducibility:** Given the same seed, the engine produces identical output on the same platform. Seeds are derived via `StdRng::seed_from_u64(seed_base + k)` for realisation  $k$ .



# Architecture

## Chapter Overview

- *Six-phase simulation pipeline*
- *Module organisation and data flow*
- *Execution modes: in-memory vs. streaming*

## 2.1 Pipeline Overview

The simulation proceeds through six phases, each corresponding to a Rust module:

|                                                                              |                                                            |   |
|------------------------------------------------------------------------------|------------------------------------------------------------|---|
| XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX |                                                            |   |
| ☒                                                                            | Phase 1: Sprinkle + Hasse (phase1/)                        | ☒ |
| ☒                                                                            | Poisson sprinkling → Transitive Reduction (CSR)            | ☒ |
| XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX |                                                            |   |
| ☒                                                                            | Phase 2: Kuratowski Contraction (phase2/)                  | ☒ |
| ☒                                                                            | $K_{\{2,N\}}$ detection → $K_5$ threat absorption → Defect | ☒ |
| XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX |                                                            |   |
| ☒                                                                            | Phase 3: Spectral Observables (phase3/)                    | ☒ |
| ☒                                                                            | Random walkers → $d_S(t)$ → Causal Flux                    | ☒ |
| XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX |                                                            |   |
| ☒                                                                            | Measurements M1-M14 (measure/)                             | ☒ |
| ☒                                                                            | M1-M10: Mass, half-life, modulo, vacuum pol.,              | ☒ |
| ☒                                                                            | electroweak, decoherence, neutrino, PMNS, Higgs,           | ☒ |
| ☒                                                                            | Lagrangian; M11-M14: Hausdorff, zigzag, collider,          | ☒ |
| ☒                                                                            | mass formula                                               | ☒ |
| XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX |                                                            |   |
| ☒                                                                            | Phase 4: Output (output/)                                  | ☒ |
| ☒                                                                            | CSV serialisation + provenance stamps                      | ☒ |
| XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX |                                                            |   |
| ☒                                                                            | Ensemble Runner (ensemble/)                                | ☒ |
| ☒                                                                            | Adaptive Welford convergence + checkpointing               | ☒ |
| XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX |                                                            |   |

## 2.2 Module Map

| Module                     | Responsibility                                                                                                                   |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <code>main.rs</code>       | CLI parsing, flag dispatch, provenance boot                                                                                      |
| <code>config.rs</code>     | TOML configuration, RAM-aware mode selection                                                                                     |
| <code>provenance.rs</code> | Five-layer SHA-256 provenance system                                                                                             |
| <code>phase1/</code>       | Poisson sprinkling ( <code>sprinkle.rs</code> ) and Hasse construction ( <code>hasse.rs</code> )                                 |
| <code>phase2/</code>       | Causal Prism detection ( <code>defect.rs</code> ), topology summary ( <code>topology.rs</code> )                                 |
| <code>phase3/</code>       | Spectral dimension ( <code>spectral.rs</code> ), random walkers ( <code>walker.rs</code> ), causal flux ( <code>flux.rs</code> ) |

|                             |                                                                                                                           |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <code>graph/</code>         | Type-safe CSR ( <i>csr.rs</i> ), grid quantisation ( <i>grid.rs</i> )                                                     |
| <code>ensemble/</code>      | Adaptive batching ( <i>runner.rs</i> ), Welford averaging ( <i>averaging.rs</i> ), checkpointing ( <i>checkpoint.rs</i> ) |
| <code>measure/</code>       | Fourteen observer measurements M1–M14, in <i>m01_traversal.rs</i> through <i>m14_mass_formula.rs</i>                      |
| <code>output/</code>        | CSV writer ( <i>csv.rs</i> ), human-readable summary ( <i>summary.rs</i> )                                                |
| <code>anim_export.rs</code> | Binary topology slice for Manim animation                                                                                 |

## 2.3 Execution Modes

### Definition 1: In-Memory Mode

All  $N$  points and the full CSR adjacency are held in RAM. This is the fast path. Memory estimate:  $N \times 2000$  bytes for  $N \leq 50,000$ ;  $N \times 500$  bytes for  $N > 50,000$  (Hasse degree bounded by 15). Selected by `--inmemory` or recommended automatically when estimated memory fits within 70% of available RAM.

### Definition 2: Streaming Mode

For extreme scales ( $N > 10^7$ ), the simulation uses time-slab sprinkles and a sliding window. Only the active window is held in RAM. Estimate:  $N \times 13 + 100$  MB overhead. Selected by `--stream`.

The mode is auto-detected by `config::recommend_mode()`, which queries `sysinfo::System` for available memory and applies the safety fraction from *causal\_set.toml*.

# Phase 1 — Poisson Sprinkling

## Chapter Overview

- *Sprinkling  $N$  events into a 4D causal diamond*
- *Volume formula and density normalisation*
- *Streaming chunks for sliding-window mode*

## 3.1 The Causal Diamond

Source: `phase1/sprinkle.rs`

The simulation domain is a 4D causal diamond: the intersection of the future light cone of a past tip  $p^-$  and the past light cone of a future tip  $p^+$ . In Cartesian coordinates, the constraint is

$$|t| + r \leq T/2, \quad r = \sqrt{x^2 + y^2 + z^2}.$$

The diamond has 4-volume  $V = \pi T^4/24$ . To normalise the Poisson density to  $\rho \approx 1$ , we set

$$T = \left( \frac{24N}{\pi} \right)^{1/4}.$$

## 3.2 Rejection Sampling

The function `sprinkle(n, rng)` draws uniform random points in the bounding box  $[-T/2, T/2]^4$  and accepts those satisfying  $|t| + r \leq T/2$ . The acceptance rate is  $\pi/24 \approx 13\%$  in 4D. The returned array has exactly  $N$  points in  $[t, x, y, z]$  format.

## 3.3 Streaming Chunks

For the sliding-window path, `sprinkle_chunk()` generates points in a time slab  $[t_{\min}, t_{\max}]$ , returning `GridPoint` structs with sequential `orig_idx` numbering. Grid quantisation fields (`qt`, `qx`, `qy`, `qz`) are filled by the caller after choosing grid dimensions.

# Phase 1 — Hasse Diagram

## Chapter Overview

- *Two-tier Hasse construction: sparse  $A^2$  vs. geometric OCI*
- *Transitive reduction as the origin of triangle-freeness*
- *Adaptive early termination valves*

## 4.1 Transitive Reduction

Source: `phase1/hasse.rs`

Given  $N$  Poisson-sprinkled events with coordinates  $p_i = (t_i, x_i, y_i, z_i)$ , two events are **causally related** iff

$$(t_j - t_i)^2 - |\vec{x}_j - \vec{x}_i|^2 > 0, \quad t_j > t_i.$$

The Hasse diagram is the **transitive reduction**: edge  $(i, j)$  exists iff  $i \prec j$  and there is no intermediate  $k$  with  $i \prec k \prec j$ . This guarantees triangle-freeness by construction — any triangle would imply a transitively redundant edge.

## 4.2 Tier 1: Sparse $A^2$ Reduction

For  $N \leq 15,000$ , the function `build_hasse_sparse()` materialises the full causal closure as a sparse matrix  $A$  (via `sprs::TriMat`), computes  $A^2$ , and retains only edges where  $A^2[i, j] = 0$  (no 2-hop path exists). This is exact but  $O(N^2)$  in space.

### Theorem 1: Sparse Reduction Correctness

In a transitively closed DAG, edge  $(i, j)$  is a Hasse link if and only if  $A^2[i, j] = 0$ .  
Proof: if a 2-hop path  $i \rightarrow k \rightarrow j$  exists, then  $j$  is reachable from  $i$  via  $k$ , making the direct edge redundant.

## 4.3 Tier 2: Geometric OCI Construction

For  $N > 15,000$ , the function `build_hasse_direct()` avoids materialising the full closure. Instead:

1. **Quantise & sort**: Coordinates are quantised to an integer lattice via `quantize_and_sort()`, yielding `GridPoint` records sorted by  $(q_t, q_x, q_y, q_z)$  for cache locality.
2. **Build OCI**: The Occupied Cell Index (OCI) is a per-layer structure mapping  $(q_x, q_y, q_z)$  cells to the points they contain, sorted by  $q_x$  for binary search.
3. **Parallel link discovery**: For each source  $u$ , iterate over  $\Delta t$  shells from 0 to `MAX_CAUSAL_DEPTH`. For each shell, query the OCI for cells within the light cone ( $r_{\max} = \Delta t + 1$ ), test causality, and check transitivity against existing children.
4. **CSR assembly**: Collect per-chunk degrees and targets, build the CSR head and data arrays, sort each neighbour list for binary-search correctness.

### 4.3.1 Adaptive Termination Valves

Three early-exit conditions accelerate the search:

1. **Degree cap** (`MAX_HASSE_DEGREE = 15`): A node with  $\geq 15$  descendants has a fully populated causal shadow. Further links are redundant. This bound also sets the *lattice*

*mode decay timescale* for spectral dimension measurements: local eigenvalues of the lazy-walk transition matrix near  $\lambda \approx 1 - 1/(2D)$  decay on a timescale  $t_{\text{lattice}} = D_{\text{max}}$ .

2. **Alexandrov volume short-circuit:** If every causal candidate in a  $\Delta t$  shell is transitively blocked (non-empty Alexandrov set between  $u$  and  $v$ ), further shells have strictly larger Alexandrov volume and will also be blocked.
3. **Consecutive dry shells:** Two consecutive shells with no causal candidates plus  $\geq 2$  existing descendants implies the light cone slice is empty.

## 4.4 Bulk Momentum

Both tiers compute the **bulk momentum** vector:  $p_i = \text{out-degree}(i) - \text{in-degree}(i)$ . This quantity encodes the net causal asymmetry at each node and is used in Phase 2 for generation classification ( $\phi(w) = \text{sign}(p_w) \in \{-1, 0, +1\}$ ).

# Graph Infrastructure

## Chapter Overview

- *Type-safe CSR with compile-time direction marker*
- *Grid quantisation and the Occupied Cell Index*
- *Physical constants encoded as Rust constants*

## 5.1 Type-Safe CSR

Source: `graph/csr.rs`

The central data structure is `CsrGraph<D>`, a Compressed Sparse Row graph parametrised by a zero-sized direction marker:

- `CsrGraph<Directed>`: forward-only edges ( $u \rightarrow v$  where  $t_u < t_v$ ). Produced by Phase 1.
- `CsrGraph<Undirected>`: both directions stored explicitly. Produced by `symmetrize()` or the defect graph.

The `PhantomData<D>` trick prevents passing a directed Hasse diagram to spectral walkers (which require the symmetric graph) at *compile time*.

### Definition 1: CSR Storage Layout

`head[u]..head[u+1]` indexes into data to give the sorted neighbour list of node  $u$ . Binary search on sorted neighbours gives  $O(\log d)$  edge queries, where  $d \leq 15$  in practice.

Key operations:

- `neighbors(u)`: returns `&[u32]` slice of sorted neighbours
- `degree(u)`: returns out-degree (or total degree for undirected)
- `has_edge(u, v)`: binary search on sorted neighbour list
- `symmetrize()`: converts `Directed`  $\rightarrow$  `Undirected` by adding reverse edges (method on `CsrGraph<Directed>` only)
- `reverse()`: reverses all edge directions (parents  $\leftrightarrow$  children)
- `unique_edges()`: extracts  $(u, v)$  pairs with  $u < v$  for eigendecomposition (method on `CsrGraph<Undirected>` only)

Free functions:

- `build_directed_csr(n, rows, cols)`: builds from unsorted edge pairs
- `build_undirected_csr(n, rows, cols)`: builds symmetric CSR from edge pairs with automatic reverse insertion

## 5.2 Grid Quantisation

Source: `graph/grid.rs`

The OCI construction requires mapping continuous coordinates to an integer lattice. The `GridPoint` struct stores both the original `[f64; 4]` coordinates and quantised (`qt`, `qx`, `qy`, `qz`) fields. The `GridDims` struct encodes lattice dimensions derived from coordinate bounds.

Key constants from `grid.rs`:

- `MAX_HASSE_DEGREE = 15`: maximum out-degree in the Hasse diagram. Also sets the

lattice mode decay timescale  $t_{\text{lattice}} = D_{\text{max}}$  for spectral dimension measurements and the CRT budget floor  $t_{\text{max}} = D_{\text{max}}$

- MAX\_CAUSAL\_DEPTH: maximum  $\Delta t$  shells to search
- MAX\_PROPER\_TIME\_SQ: proper-time cutoff for causality

# Phase 2 — Kuratowski Contraction

## Chapter Overview

- Causal Prism ( $K_{2,N}$ ) detection via 2-hop traversal
- $K_5$  threat contraction for planarity preservation
- Generation classification by bulk-momentum phase signature
- Phase-coherence decomposition: visible, dark, sterile mass

## 6.1 Causal Prism Detection

Source: `phase2/defect.rs`

The Causal Prism is the fundamental topological particle in FEG. It is a complete bipartite subgraph  $K_{2,N}$  embedded in the triangle-free Hasse DAG:

### Definition 1: Causal Prism

A **Causal Prism** consists of:

- An **origin**  $u$  (past pole) and **destination**  $v$  (future pole) at graph-distance  $\geq 2$
- $N \geq 3$  **intermediaries**  $w_1, \dots, w_N$  such that  $u \rightarrow w_i$  and  $w_i \rightarrow v$  for all  $i$
- The intermediaries are mutually disconnected (guaranteed by the triangle-free property)

The minimum  $N = 3$  (constant `MIN_PRISM_SHARED`) derives from the Uniqueness of Bifurcation-Convergence theorem:  $K_{2,2}$  yields only  $S_2 \rightarrow U(1)$ ; the  $SU(3)$  gauge structure demands  $S_3$ , hence  $N \geq 3$ .

### 6.1.1 Detection Algorithm

The 2-hop forward-forward traversal:

1. Select a **core** of the top 10% of nodes by undirected degree (out-degree + in-degree from the reverse CSR).
2. For each core node  $u$  (not yet placed):
  - (a) Collect 2-hop candidates: for each child  $w$  of  $u$ , for each child  $v$  of  $w$  that is also in the core and not yet placed, add  $v$  to the candidate set.
  - (b) For each candidate  $v$ : compute  $\text{belly} = \text{children}(u) \cap \text{parents}(v)$  using sorted-slice intersection ( $O(d_u + d_v)$ ).
  - (c) Select the best partner  $v^*$  with maximum  $|\text{belly}|$ .
  - (d) If  $|\text{belly}| \geq 3$ : commit the prism, mark all component nodes as placed.

### Key Result

The algorithm is  $O(N \cdot D^2)$  where  $D = \max \text{Hasse degree} \leq 15$ . Since  $D = O(1)$ , the total complexity is  $O(N)$ .

## 6.2 $K_5$ Threat Contraction

After prism detection, external nodes threatening to form a  $K_5$  minor are absorbed:



**Definition 2:  $K_5$  Threat Criterion**

An external node  $t$  is a **threat** to prism  $(u, v, \{w_i\})$  if:

- $t$  is connected (undirected) to *both* poles  $u$  and  $v$
- $t$  is connected to  $\geq 2$  intermediaries (constant PRISM\_THREAT)

The five nodes  $\{u, v, t, w_i, w_j\}$  are then mutually reachable, forming a  $K_5$  minor.

Contraction:  $t$  is merged into the higher-degree pole.

Merge chains are resolved transitively: if  $t_1 \rightarrow t_2 \rightarrow u$ , then  $t_1$  maps directly to  $u$ . The resulting merge\_map is a forest of stars.

## 6.3 Generation Classification

Each intermediate  $w_i$  carries a **causal phase**  $\phi(w_i) = \text{sign}(\text{bulk\_momentum}[w_i]) \in \{-1, 0, +1\}$ .

**Definition 3: Generation Number**

The **generation** of a prism  $P$  is  $g(P) = |\{\phi(w_i) : w_i \in \text{intermediaries}\}|$ , i.e., the number of distinct phase classes. Since  $\phi \in \{-1, 0, +1\}$ ,  $g \in \{1, 2, 3\}$ .

**Definition 4: Net Phase and Matter/Antimatter**

The **net phase**  $\Phi(P) = \sum_i \phi(w_i)$ .

- $\Phi > 0$ : **matter** (e.g., electron for  $g = 1$ )
- $\Phi < 0$ : **antimatter** (e.g., positron for  $g = 1$ )
- $\Phi = 0$ : **sterile** (dark matter candidate)

The generation sets (GenerationSets) store node indices for Gen1, Gen2, Gen3, Anti-Gen1, and Sterile prisms. Average topological mass per generation is  $\langle M \rangle_g = \langle N \rangle$  (average intermediate count).

## 6.4 Phase-Coherence Decomposition

The defect module computes a decomposition of gravitational mass into visible and dark components:

$$M_{\text{grav}}(P) = N, \quad (6.1)$$

$$M_{\text{vis}}(P) = |\Phi(P)|, \quad (6.2)$$

$$M_{\text{dark}}(P) = N - |\Phi(P)|. \quad (6.3)$$

Aggregated ratios:

- $\Omega_{\text{dark}}/\Omega_{\text{vis}} = \sum(N - |\Phi|)/\sum|\Phi|$
- $Q_{\text{topo}} = 1/4$  (exact, from topological collider M13)
- $\alpha_0 = Q_{\text{topo}}/(8\pi) = 1/(32\pi) \approx 1/100.5$  (bare Planck coupling; vacuum polarization screens to  $\alpha \approx 1/137$ )

## 6.5 Topology Summary

Source: `phase2/topology.rs`

The TopologySummary struct aggregates 27 fields including total nodes/prisms, per-generation counts and masses, phase census,  $Q_{\text{topo}}$ ,  $\alpha_{\text{EM}}$ , and the prism histogram (belly-size distribution).

# Phase 3 — Spectral Observables

## Chapter Overview

- Spectral dimension  $d_S(t)$  from random walks
- Two computation tiers: eigendecomposition vs. Monte Carlo walkers
- Causal flux: attraction and repulsion between generations

## 7.1 Spectral Dimension

Source: `phase3/spectral.rs`

The spectral dimension quantifies the effective dimensionality experienced by a diffusing particle at scale  $t$ :

$$d_S(t) = -2 \frac{d \ln P(t)}{d \ln t},$$

where  $P(t)$  is the return probability after  $t$  lazy random walk steps on the graph. Centred finite differences are used for interior points; forward/backward differences at endpoints.

The lazy walk (50% stay, 50% hop to a random neighbour) has transition matrix

$$T = \frac{1}{2}(I + D^{-1}A),$$

where  $A$  is the adjacency matrix and  $D$  the degree matrix. Eigenvalues of  $T$  lie in  $[0, 1]$  with  $\lambda_0 = 1$  for the stationary mode. The return probability decomposes as

$$P(t) = \frac{1}{N} \text{Tr}(T^t) = \frac{1}{N} \sum_k \lambda_k^t.$$

Three eigenvalue bands contribute at different timescales:

| Band      | Eigenvalues                         | Decay timescale       |
|-----------|-------------------------------------|-----------------------|
| Continuum | $\lambda \approx 1$                 | $O(N^{2/d_S})$        |
| Lattice   | $\lambda \approx 1 - O(1/D_{\max})$ | $O(D_{\max}) = O(15)$ |
| High-freq | $\lambda \approx 0$                 | $O(1)$                |

After  $t \approx D_{\max} = 15$  steps, the lattice modes have decayed by  $\sim 1/e$  and  $d_S(t)$  enters the continuum scaling plateau (the **lattice-to-continuum crossover**). This is *not* a mixing time in the coupon-collector sense (that would be  $O(D \ln D) \approx 41$ ) — it is the spectral decay timescale on which discrete graph artifacts (degree echoes, parity oscillations) damp below the continuum signal.

### Definition 1: WalkResult

Each walk category (vacuum, defect, per-generation, sterile, flux) returns:

- `p`:  $P(t)$  return probability at each measurement step
- `ds`:  $d_S(t)$  spectral dimension derived from  $P(t)$
- `ds_std`: ensemble standard deviation (empty for  $M = 1$ )

**Definition 2: SpectralOutput**

The composed output struct groups:

- Vacuum (global + core) and defect (global + core) walks
- Per-generation walks: [Gen1, Gen2, Gen3, Anti1]
- Sterile walk (phase-cancelled prisms)
- Causal flux: attraction (Gen1→Anti1) and repulsion (Gen1→Gen1)
- Normalised flux: flux per target node
- Average topological mass per generation

## 7.2 Tier 1: Eigendecomposition

For  $N \leq \text{eigen\_cutoff}$  (default 3000), the function `compute_eigen()` computes the full eigendecomposition of the symmetric normalised adjacency  $S = D^{-1/2}AD^{-1/2}$  via nalgebra. The return probability is then

$$P(t) = \frac{1}{N} \sum_{k=0}^{N-1} \lambda_k^t,$$

where  $\lambda_k$  are the eigenvalues of  $S$  (clamped to  $[-1, 1]$ ). Core walks restrict the sum to eigenvector-weighted contributions on core nodes.

## 7.3 Tier 2: Monte Carlo Walkers

Source: `phase3/walker.rs`

For  $N > \text{eigen\_cutoff}$ , lazy random walkers simulate diffusion. Each walker implements the transition matrix  $T = \frac{1}{2}(I + D^{-1}A)$  stochastically: at each step it stays with probability  $\frac{1}{2}$  or hops to a uniformly random neighbour with probability  $\frac{1}{2}$ . The return indicator  $X_i \in \{0, 1\}$  at each measurement step is a Bernoulli( $P(t)$ ) variable. By the CLT,  $\text{RSE} = 1/\sqrt{W \cdot P(t)}$ .

The number of walkers  $W$  is set by the **Causal Resolution Theorem**:

$$W = \lceil (t_{\max}/\varepsilon)^2 \rceil,$$

where  $\varepsilon$  is the topological error tolerance (default 0.01) and  $t_{\max}$  is the maximum diffusion time (default 15 =  $D_{\max}$ ). Setting  $\text{RSE} < \varepsilon$  with  $P(t) \sim t^{-2}$  ( $d_S = 4$ ) yields  $W > t^2/\varepsilon^2$ . At  $\varepsilon = 0.01$ ,  $t_{\max} = 15$ :  $W = 2,250,000$  walkers.

**Why  $t_{\max} = D_{\max} = 15$ :** This is the *lattice mode decay timescale* — after  $\sim D_{\max}$  steps, the discrete graph artifacts in  $P(t)$  have damped by  $\sim 1/e$  and the continuum scaling plateau begins. Calibrating at the plateau onset is optimal because  $P(D_{\max})$  is the largest return probability in the scaling regime, yielding the smallest required  $W$ . For  $t > D_{\max}$ , RSE degrades as  $(t/D_{\max}) \cdot \varepsilon$ , but auto-convergence (Welford RSE  $< 5 \times 10^{-4}$ ,  $k = 3$  consecutive batches) adds walker batches until the observable at  $t \approx 36$  (midpoint of the step array, well inside the scaling plateau) stabilises.

Return counts are accumulated as u64 integers (strict finitism).

## 7.4 Causal Flux

Source: `phase3/flux.rs`

The causal flux measures directed transmission between generation sets:

- **Attraction** (Gen1→AntiGen1): flux from matter to antimatter (opposite topological charge → attraction)
- **Repulsion** (Gen1→Gen1): flux from matter to matter (same charge → repulsion)

Normalised flux divides by the number of target nodes, isolating the intrinsic coupling per unit charge.

## 7.5 Diffusion Steps

The measurement times are densely sampled near  $t = 1$ :

```
steps = (1..=30).chain((16..=50).map(|i| i * 2))
```

giving steps 1–30 (unit spacing) and 32, 34, ..., 100 (stride 2). This covers three regimes:

- $t \lesssim D_{\max} = 15$ : short-time lattice transient (degree echoes, parity oscillations from discrete graph structure)
- $D_{\max} \lesssim t \lesssim N^{2/d_S}$ : continuum scaling plateau where  $P(t) \sim t^{-d_S/2}$  and  $d_S(t)$  is physically meaningful
- $t \gtrsim N^{2/d_S}$ : approach to finite-size saturation  $P(t) \rightarrow 1/N$

The convergence observable is  $d_S$  at the midpoint of the step array ( $t \approx 36$ ), which lies well within the scaling plateau.

# Ensemble Framework

## Chapter Overview

- *Adaptive batching with Welford convergence*
- *Checkpointing and resume capability*
- *Seed derivation for deterministic reproducibility*

## 8.1 Ensemble Runner

Source: `ensemble/runner.rs`

Each ensemble run consists of  $M$  independent Monte Carlo realisations of the full Phase 1–3 pipeline. The runner uses **adaptive Welford convergence**:

1. Run `min_ensemble` realisations (default 8) unconditionally.
2. After each subsequent batch, compute the online running mean and standard error of `mass_gen1` via the Welford algorithm.
3. If the relative standard error  $SE/\bar{x} < \text{target\_error}$  (default 0.01), declare convergence and stop.
4. Otherwise, continue up to `max_ensemble` (default 50).

### Definition 1: EnsembleResult

The output contains:

- `spectral`: Ensemble-averaged `SpectralOutput`
- `topology`: Aggregated `TopologySummary`
- `measurements`: Aggregated `MeasureResults` (if any active)
- `actual_m`: Number of realisations actually run
- `converged`: Whether the target error was reached

## 8.2 Seed Derivation

Realisation  $k$  uses seed `seed_base + k`. The base seed is either user-specified (`--seed`) or derived from the system clock. The deterministic seeding ensures reproducibility: given the same (`seed`,  $N$ ,  $M$ ), the output is identical on the same platform.

## 8.3 Concurrency

Realisations within a batch run in parallel via `rayon`. The maximum concurrency is:

- `min(batch_size, max_concurrent_runs)`
- `max_concurrent_runs`: for  $N > 10^7$ , limited to 2; otherwise 4
- Overridable via `--threads` and `--batch-size`

## 8.4 Checkpointing

Source: `ensemble/checkpoint.rs`

After each realisation, `spectral` and `topology` results are serialised to a checkpoint file in the output directory. The `--resume` flag reloads the checkpoint and continues from the last completed realisation. Provenance verification prevents cross-fork resume (Layer 4 of the provenance system).

## 8.5 Averaging

Source: *ensemble/averaging.rs*

The `average_ensemble()` function computes element-wise means of  $P(t)$  and  $d_S(t)$  vectors across  $M$  realisations, plus ensemble standard deviations. Topology fields are aggregated by `aggregate_topology()`.

# Measurements M1–M5

## Chapter Overview

- *M1: Traversal mass ratios*
- *M2: Half-life census*
- *M3: Modulo path integral (NTT)*
- *M4: Vacuum polarisation*
- *M5: Electroweak sector*

Source: `measure/mod.rs`

The measurement system provides fourteen independent physics observables. Each measurement is gated by a boolean flag in `MeasureFlags` and receives a `MeasureContext` containing the full Phase 1–2 outputs. Results are aggregated across ensemble realisations via per-measurement `aggregate()` functions.

## 9.1 M1: Traversal Mass Ratios

Source: `measure/m01_traversal.rs`

Flag: `--measure-mass`

Measures the **topological inertia** of Causal Prisms by confining random walkers to prism components and recording cover times. The ratio of cover times between generations gives the mass hierarchy:

$$\frac{m_\mu}{m_e} \approx \frac{\langle T_{\text{cover}} \rangle_{\text{Gen2}}}{\langle T_{\text{cover}} \rangle_{\text{Gen1}}}.$$

Output: `traversal_mass.csv` with per-generation cover times and ratios.

## 9.2 M2: Half-Life Census

Source: `measure/m02_halflife.rs`

Flag: `--measure-halflife`

Measures the topological stability of prisms by recording edge-snap events — the random removal of one intermediary edge that splits a  $K_{2,N}$  into  $K_{2,N-1} + K_{2,1}$ . The half-life is the Markov hitting time for the first snap event at a given thermal excitation.

### Key Result

*The electron ( $K_{2,3}$ ) is topologically stable: splitting to  $K_{2,2} + K_{2,1}$  requires overcoming the  $K_5$  planarity barrier (energy  $\sim M_{Pl}$ ). The muon and tau have finite half-lives proportional to their genus gap.*

Output: `half_life.csv` with per-generation half-lives and snap rates.

## 9.3 M3: Modulo Path Integral (NTT)

Source: `measure/m03_modulo.rs`

Flag: `--measure-modulo`

Implements the **Number-Theoretic Transform** (NTT) path integral: walkers accumulate action  $S$  as  $g^S \bmod p$  where  $g$  is a primitive root and  $p$  is a prime modulus. This replaces the continuum  $e^{iS/\hbar}$  with a discrete  $\mathbb{F}_p$  phase.

Configuration:

- prime: default  $65537 = 2^{16} + 1$  (Fermat prime)
- root: default 3 (primitive root of 65537)
- steps: default 500 NTT walk steps

The interference pattern emerges from modular cancellation: when  $\sum g^{S_j} \equiv 0 \pmod{p}$ , the path is destructively forbidden.

Output: *modulo\_interference.csv* with phase histograms and interference fringe visibility.

## 9.4 M4: Vacuum Polarisation

Source: *measure/m04\_vacuum\_pol.rs*

Flag: `--measure-vacuum`

Measures the running of the topological charge  $Q_{\text{topo}}$  with scale. At short distances (small walker steps), the effective coupling increases due to virtual prism–anti-prism screening:

$$\alpha_{\text{EM}}^{-1}(0) \approx 132 \quad \rightarrow \quad \alpha_{\text{EM}}^{-1}(m_Z) \approx 128.$$

The mechanism is planarity screening: at short distances, prism–anti-prism pairs are resolved and screen the effective charge.

Output: *vacuum\_polarization.csv* with  $Q_{\text{topo}}(t)$  and  $\alpha^{-1}(t)$  at each scale  $t$ .

## 9.5 M5: Electroweak Sector

Source: *measure/m05\_electroweak.rs*

Flag: `--measure-electroweak`

Probes the separation of  $SU(2)_L \times U(1)_Y$  from the prism topology. This measurement uses the asymmetry between left-handed and right-handed walker trajectories to extract the Weinberg angle  $\theta_W$ :

$$\sin^2 \theta_W = \frac{\text{left-blocked}}{\text{left-blocked} + \text{right-blocked}}.$$

Output: *electroweak.csv* with mixing angle and left/right asymmetry.



# Measurements M6–M10

## Chapter Overview

- *M6: Quantum decoherence and Born rule*
- *M7: Neutrino census*
- *M8: PMNS mixing matrix*
- *M9: Higgs mechanism (topological drag)*
- *M10: Full Standard Model Lagrangian card*

## 10.1 M6: Quantum Decoherence

Source: `measure/m06_decoherence.rs`

Flag: `--measure-decoherence`

Tests the emergence of the Born rule via the Cramer transactional handshake. A retarded wave (fwd) is seeded from prism intermediates and propagated forward with NTT phase rotation  $W = g^{(p-1)/4} \bmod p$  (90° per edge); an advanced wave (bwd) is seeded from the future boundary and propagated backward with  $W^{-1}$ . The handshake product, centered symmetrically in  $\mathbb{Z}/p\mathbb{Z}$ , gives the Born amplitude at each environment node:

$$P(d) \propto |\text{fwd}(d) \times \text{bwd}(d)|^2.$$

$K_{2,2}$  diamond counts at environment nodes provide the observed detection events. Pearson  $r$  between the predicted  $|\psi|^2$  PMF and the observed  $K_{2,2}$  PMF, validated against a permutation null model (1000 shuffles), measures Born rule compliance.

Output: `decoherence.csv` and `born_rule.csv` with Pearson  $r$ , null model statistics, and per-bin predicted vs. observed frequencies.

## 10.2 M7: Neutrino Census

Source: `measure/m07_neutrino.rs`

Flag: `--measure-neutrino`

Counts “open prisms” —  $K_{2,1}$  structures (two poles, one intermediary) that are topologically sterile ( $\Phi = 0$ ) and have near-zero topological mass. These are identified as neutrinos: gravitationally active but electromagnetically invisible.

The census provides per-generation neutrino counts and the neutrino/charged-lepton ratio.

Output: `neutrino.csv` with generation-resolved neutrino counts.

## 10.3 M8: PMNS Mixing Matrix

Source: `measure/m08_pmns.rs`

Flag: `--measure-pmns`

Requires M7 (automatically enabled). Extracts the PMNS mixing matrix from neutrino oscillation probabilities measured via walker transmission between generation-tagged open prisms. The mixing angles ( $\theta_{12}, \theta_{13}, \theta_{23}$ ) are extracted from the matrix elements.

**Key Result**

*FEG predicts “anarchy”-style PMNS mixing: large mixing angles emerging from the random topology of the Hasse diagram, in agreement with the observed near-maximal  $\theta_{23}$  and large  $\theta_{12}$ .*

Output: `pmns.csv` with the  $3 \times 3$  mixing matrix and extracted angles.

## 10.4 M9: Higgs Mechanism

Source: `measure/m09_higgs.rs`

Flag: `--measure-higgs`

Models the Higgs boson as a “breathing mode” of the Causal Prism ensemble: a coherent oscillation of the average belly size  $\langle N \rangle$  about its equilibrium value. The entropy cost  $\Delta S \sim 1/\alpha$  of a uniform shift in  $N$  sets the Higgs mass scale:

$$m_H \sim v\sqrt{2\lambda}, \quad \lambda \sim \alpha^2.$$

The measurement applies topological drag — walkers on prism components experience an effective potential proportional to their belly size, and the oscillation frequency is extracted from the autocorrelation.

Output: `higgs.csv` with breathing-mode frequency, effective  $\lambda$ , and mass estimate.

## 10.5 M10: Standard Model Lagrangian Card

Source: `measure/m10_lagrangian.rs`

Flag: `--measure-lagrangian`

Assembles a complete Standard Model parameter card from M1–M9 outputs:

| Parameter              | Source | FEG Derivation                               |
|------------------------|--------|----------------------------------------------|
| $\alpha_0 = 1/(32\pi)$ | M13    | Topological collider $Q_{\text{topo}} = 1/4$ |
| $\sin^2 \theta_W$      | M5     | Left/right walker asymmetry                  |
| $m_e : m_\mu : m_\tau$ | M1     | Cover-time ratios                            |
| PMNS matrix            | M8     | Neutrino walker transmission                 |
| $m_H$                  | M9     | Breathing-mode frequency                     |
| $\tau_\mu, \tau_\tau$  | M2     | Edge-snap hitting times                      |
| $\alpha_s$             | M4     | Short-distance $Q_{\text{topo}}$ running     |
| $P =  K ^2$            | M6     | Modular path integral                        |
| Neutrino counts        | M7     | Open-prism census                            |

This is a post-hoc assembly: M10 does not run its own simulation, but collects and cross-validates the outputs of M1–M9.

Output: `lagrangian.csv` with the full parameter card.

# Measurements M11–M14

## Chapter Overview

- *M11: Hausdorff dimension via BFS volume growth*
- *M12: Zigzag Kaluza–Klein dimension*
- *M13: Topological particle collider ( $Q_{\text{topo}} = 1/4$ )*
- *M14: Exact mass formula (genus ladder)*

These four measurements extract deterministic topological observables that do not require Phase 3 random walks. They are automatically enabled by `--topology-only` mode and are also included in `--measure-all`.

## 11.1 M11: Hausdorff Dimension

Source: `measure/m11_hausdorff.rs`

Flag: `--measure-hausdorff`

Measures the Hausdorff dimension  $d_H$  by sampling random source nodes and performing BFS on both the directed (forward-only) and undirected (symmetric) vacuum CSR. Volume growth  $V(r) \sim r^{d_H}$  is fitted via log-log linear regression over the power-law region ( $r = 1 \dots 8$ , before saturation):

$$d_H = \frac{d \ln V}{d \ln r} \approx 4.0 \quad (\text{directed BFS}).$$

The spectral dimension  $d_S \approx 2$  at UV is a lazy-walk artifact on the tree-like Hasse DAG; Hausdorff dimension via volume growth is the correct geometric observable for the causal set.

### Key Result

At  $N = 10^7$ ,  $M = 1$ :  $d_H(\text{directed}) = 4.0$  (single realisation; ensemble average converges to 4.0 exactly).

Output: `hausdorff.csv` with  $V(r)$  at each BFS depth and fitted  $d_H$ .

## 11.2 M12: Zigzag KK Dimension

Source: `measure/m12_zigzag.rs`

Flag: `--measure-zigzag`

Measures the zigzag Kaluza–Klein dimension  $d_{\text{zig}}$  by alternating forward and backward BFS steps on the directed vacuum CSR and its reverse. The total volume growth  $V(r)$  is fitted as  $V \sim r^{d_{\text{zig}}}$ .

The zigzag path explores both future and past causal directions, effectively probing the dimensionality of the full causal structure including KK-like extra directions:

$$d_{\text{zig}} \approx 4.004.$$

The screening ratio  $S_4/S_{d_{\text{zig}}} = 32\pi/137 \approx 0.733$  connects the bare coupling  $\alpha_0 = 1/(32\pi)$  to the observed  $\alpha \approx 1/137$ .

Output: `zigzag.csv` with zigzag  $V(r)$  and fitted  $d_{\text{zig}}$ .

### 11.3 M13: Topological Particle Collider

Source: `measure/m13_collider.rs`

Flag: `--measure-collider`

Extracts  $Q_{\text{topo}}$  from a strict bipartite seam plus external Markov blanket analysis of  $K_{2,N}$  prisms. For each committed prism:

1. **Seam**: the exclusive children of each pole, filtered by  $z > \text{emitting\_top}$  (no tachyons)
2. **Blanket**: edges from diamond nodes to non-diamond nodes only (strict external boundary)
3.  $Q_{\text{topo}} = \text{seam\_size}/\text{blanket\_size}$

#### Key Result

$Q_{\text{topo}} = 1/4$  (exact, locked asymptote across  $N = 10^4$  to  $10^7$ ). The bare coupling is  $\alpha_0 = Q_{\text{topo}}/(8\pi) = 1/(32\pi) \approx 1/100.5$ . This supersedes the earlier “port counting” method ( $Q_{\text{topo}} = 4/21$ ) used in earlier versions.

Output: `collider.csv` with per-prism  $Q_{\text{topo}}$  and ensemble mean.

### 11.4 M14: Exact Mass Formula

Source: `measure/m14_mass_formula.rs`

Flag: `--measure-mass-formula`

Computes the genus  $g$  of every  $K_{2,n}$  prism via the Grothendieck–Euler formula (`phase2/writhe.rs`) and verifies the exact mass formula:

$$M(g) = M_e \cdot 2^g \cdot (32\pi)^g / \binom{\max(3, 2g+1)}{3}.$$

Three factors:

- $2^g$ : parity filter (binary Kuratowski junction per hole)
- $(32\pi)^g = (1/\alpha_0)^g$ : parallel homology drag (metric wrap per hole)
- $\binom{n_{\min}}{3}$ : subgraph superposition ( $K_{2,3}$  backbone count in  $K_{2,n}$ ), where  $n_{\min}$  from the Grothendieck–Euler floor  $F = n - 2g \geq 1$

#### Key Result

Zero free parameters:

- $m_\mu/m_e = 201.06$  (SM: 206.77,  $-2.76\%$ )
- $m_\tau/m_e = 4042.6$  (SM: 3477.5,  $+16.2\%$ )

Genus classification:  $g = 0 \rightarrow \text{electron}$ ,  $g = 1 \rightarrow \text{muon}$ ,  $g = 2 \rightarrow \text{tau}$ .

Output: `mass_formula.csv` with genus histogram, mean belly per genus, and predicted mass ratios.

# Output & CSV

## Chapter Overview

- *CSV serialisation with provenance headers*
- *36-column spectral CSV format*
- *Topology summary and mass spectrum CSVs*
- *Human-readable terminal summary*

## 12.1 CSV Writer

Source: `output/csv.rs`

The `CsvWriter` struct wraps a `BufWriter<File>` and prepends the standard provenance header block (Layer 3):

```
# FEG Kuratowski Calculus Engine (FEG_prism)
# Author: Juan Pablo Silva Alvarado
# DOI: 10.5281/zenodo.18769707
# Framework: Fractal Entropic Geometrodynamics
# Provenance SHA-256: a4fee337...
# Commit: 5b07174 (dirty)
# Timestamp: 2026-02-24T12:00:00Z
```

## 12.2 Spectral CSV

Source: `output/mod.rs`

The function `write_spectral_csv()` produces `results.csv` with 36 columns per diffusion step:

| Column(s)                      | Description                                      |
|--------------------------------|--------------------------------------------------|
| step                           | Diffusion time $t$ (integer steps)               |
| P_vac, dS_vac                  | Vacuum return probability and spectral dimension |
| P_def, dS_def                  | Defect (post-contraction) return prob. and $d_S$ |
| P_Gen1..3, dS_Gen1..3          | Per-generation walks                             |
| P_Anti1, dS_Anti1              | Antimatter generation walk                       |
| Flux_Attr, Flux_Repu           | Causal flux (raw)                                |
| Flux_Attr_Norm, Flux_Repu_Norm | Normalised flux per target                       |
| P_Sterile, dS_Sterile          | Sterile prism walk                               |
| Mass_Gen1..3, Mass_Anti1       | Average topological mass                         |
| dS*_std                        | Ensemble standard deviations                     |
| Flux*_std                      | Flux ensemble standard deviations                |

## 12.3 Topology CSV

The function `write_topology_csv()` writes `topology_summary.csv` with all 27 `TopologySummary` fields. The function `write_mass_spectrum_csv()` writes `mass_spectrum.csv` with the prism histogram (belly size  $\rightarrow$  count).

## 12.4 Measurement CSVs

The function `measure::write_all_csv()` dispatches to per-measurement `write_csv()` functions. Output files:

- *traversal\_mass.csv* (M1)
- *half\_life.csv* (M2)
- *modulo\_interference.csv* (M3)
- *vacuum\_polarization.csv* (M4)
- *electroweak.csv* (M5)
- *decoherence.csv*, *born\_rule.csv* (M6)
- *neutrino.csv* (M7)
- *pmns.csv* (M8)
- *higgs.csv* (M9)
- *lagrangian.csv* (M10)
- *hausdorff.csv* (M11)
- *zigzag.csv* (M12)
- *collider.csv* (M13)
- *mass\_formula.csv* (M14)

## 12.5 Terminal Summary

Source: *output/summary.rs*

The `print_summary()` function writes a human-readable report to stdout including: spectral dimension at key scales, mass hierarchy ratios, prism counts,  $\alpha_{\text{EM}}$ , dark matter fraction, and measurement highlights.

## 12.6 Metadata

The Metadata struct accompanies every CSV file:

- `n_points`, `actual_m`, `converged`
- `min_ensemble`, `max_ensemble`
- `mode` (“in-memory” or “streaming”)
- `epsilon`, `tmax`, `walkers`
- `seed`, `timestamp`, `commit`

# Animation Export

## Chapter Overview

- *Binary topology slice for Manim scenes*
- *Compact bincode serialisation*
- *Integration with the animation layer*

## 13.1 Topology Slice

Source: `anim_export.rs`

The animation export produces a single-realisation snapshot containing:

### Definition 1: TopologySlice

- `n_total`: number of spacetime events
- `coordinates`:  $[t, x, y, z]$  for each event
- `hasse_edges`: directed  $(u, v)$  Hasse links
- `prisms`: vector of `PrismDef` (origin, destination, intermediates)

The slice is serialised via bincode to a compact binary file. Typical size:  $\sim 1$  MB for  $N = 5000$ .

## 13.2 Usage

Invoked by the `--export-slice` flag:

```
feg_prism 5000 1 ./output --inmemory \
  --export-slice ./output/slice.bin --seed 42
```

This runs a single realisation (Phases 1–2 only, no spectral walkers) and writes the topology slice. The Python animation layer (`animations/causal_anim/scene.py`) loads the slice and renders:

- 3D Hasse diagram with prism highlighting
- Time-evolution of causal structure
- Generation-coloured prism components

# Configuration

## Chapter Overview

- TOML configuration file `causal_set.toml`
- RAM-aware mode selection
- Memory estimation formulas

## 14.1 Configuration File

Source: `config.rs`

The engine reads `causal_set.toml` from the output directory or current directory. If absent, a default is created:

```
# FEG Prism Simulation - configuration
output_dir = "."
max_ram_gb = 0          # 0 = auto-detect
safety_fraction = 0.70
```

## 14.2 Config Struct

### Definition 1: Config Fields

- `max_ram_bytes`: hard RAM limit (0 = use system detection)
- `safety_fraction`: fraction of available RAM considered safe (default 0.70, clamped to [0.05, 1.0])
- `output_dir`: default output directory (CLI overrides)

## 14.3 Memory Estimation

### Definition 2: Memory Formulas

In-memory mode:

$$\text{est}(N) = \begin{cases} N^2 \times 24 & N \leq 3000 \text{ (eigendecomp: dense matrix)} \\ N \times 2000 & 3000 < N \leq 50,000 \\ N \times 500 & N > 50,000 \text{ (Hasse degree } \leq 15) \end{cases}$$

Streaming mode:

$$\text{est}(N) = N \times 13 + 100 \text{ MB}$$

The recommendation engine (`recommend_mode()`) computes  $\text{est}(N) \times \text{concurrent}$  and compares to  $\text{available RAM} \times \text{safety\_fraction}$ .

## 14.4 Mode Selection Flow

1. If `--inmemory`: force in-memory.
2. If `--stream`: force streaming.
3. Otherwise: auto-detect via `recommend_mode()`. Display memory summary and prompt user to confirm or override.



# Provenance

## Chapter Overview

- *Five-layer SHA-256 provenance system*
- *No external crate for SHA-256 (custom implementation)*
- *Binding the binary to its author, DOI, and git commit*

## 15.1 Architecture

Source: `provenance.rs`

The provenance system provides a chain of trust from source code to output files:

| Layer | When    | Purpose                                                            |
|-------|---------|--------------------------------------------------------------------|
| 1     | Compile | <code>#[test]</code> canary: <code>sha256(PREIMAGE) == HASH</code> |
| 2     | Startup | Runtime verification: <code>verify_provenance()</code>             |
| 3     | Output  | CSV header stamps with author, DOI, hash, commit                   |
| 4     | Resume  | Checkpoint provenance: reject cross-fork resume                    |
| 5     | Build   | <code>env!()</code> embedding: git hash + dirty flag               |

## 15.2 Preimage and Hash

The provenance preimage is the ASCII string:

```
Juan Pablo Silva Alvarado:10.5281/zenodo.18733424:
FEG-Kuratowski-2026
```

Its SHA-256 hash is:

```
a4fee33737305b5eb78a16be2ff233bd
ce125bffa352127fcbcd997d018c05aa
```

Both the preimage and hash are compiled into the binary as constants.

## 15.3 Custom SHA-256

The engine implements SHA-256 from scratch without external crates (204 lines). The implementation:

- Supports messages up to 119 bytes (sufficient for the preimage)
- Follows FIPS 180-4 exactly: padding, schedule expansion, 64-round compression with the standard  $K$  constants
- Verified by the compile-time canary and two additional unit tests (`hex_matches_bytes`, `runtime_verify`)

## 15.4 Build-Time Embedding

The `build.rs` script (not shown) sets two environment variables:

- `GIT_HASH`: short git commit hash
- `GIT_DIRTY`: “dirty” or “clean”

These are embedded via `env!()` macros and surfaced by `commit_string()`: e.g., `"5b07174 (dirty)"`.

## 15.5 UTC Timestamp

The function `utc_timestamp()` computes the current time without external crates, using `SystemTime::now()` and manual calendar arithmetic (leap-year-aware). Format: ISO 8601 (2026-02-24T12:00:00Z).

# Usage & CLI Reference

## Chapter Overview

- Complete CLI reference with all flags
- Quick-start examples
- Common workflows

## 16.1 Synopsis

```
feg_prism [N] [M] [output_dir] [FLAGS...]
```

### 16.1.1 Positional Arguments

| Argument   | Default | Description                                                         |
|------------|---------|---------------------------------------------------------------------|
| N          | 5000    | Number of spacetime events                                          |
| M          | 50      | Maximum ensemble realisations (or use <code>--max-ensemble</code> ) |
| output_dir | .       | Directory for all output files                                      |

## 16.2 Flags

### 16.2.1 Execution Mode

| Flag                    | Description                    |
|-------------------------|--------------------------------|
| <code>--inmemory</code> | Force in-memory mode (fast)    |
| <code>--stream</code>   | Force streaming mode (low RAM) |
| <code>--resume</code>   | Resume from last checkpoint    |

### 16.2.2 Physics Parameters

| Flag                                  | Default | Description                                                               |
|---------------------------------------|---------|---------------------------------------------------------------------------|
| <code>--epsilon &lt;f64&gt;</code>    | 0.01    | Topological error tolerance $\varepsilon$                                 |
| <code>--tmax &lt;usize&gt;</code>     | 15      | CRT budget calibration time (= $D_{\max}$ , lattice mode decay timescale) |
| <code>--seed &lt;u64&gt;</code>       | clock   | Base seed (deterministic mode)                                            |
| <code>--eigen-cutoff &lt;N&gt;</code> | 3000    | Eigendecomp threshold                                                     |

### 16.2.3 Ensemble Control

| Flag                                  | Default | Description                      |
|---------------------------------------|---------|----------------------------------|
| <code>--max-ensemble &lt;N&gt;</code> | 50      | Safety cap on realisations       |
| <code>--min-ensemble &lt;N&gt;</code> | 8       | Minimum before convergence check |
| <code>--target-error &lt;F&gt;</code> | 0.01    | Relative SE target               |
| <code>--batch-size &lt;N&gt;</code>   | 4       | Max concurrent per batch         |

---

```
--threads <N>      auto      Max parallel realisations
```

---

## 16.2.4 Measurements

| Flag                   | Measurement                                                          |
|------------------------|----------------------------------------------------------------------|
| --measure-mass         | M1: Traversal mass ratios                                            |
| --measure-halflife     | M2: Half-life census                                                 |
| --measure-modulo       | M3: Modulo path integral                                             |
| --measure-vacuum       | M4: Vacuum polarisation                                              |
| --measure-electroweak  | M5: Electroweak sector                                               |
| --measure-decoherence  | M6: Quantum decoherence                                              |
| --measure-neutrino     | M7: Neutrino census                                                  |
| --measure-pmns         | M8: PMNS mixing matrix (enables M7)                                  |
| --measure-higgs        | M9: Higgs mechanism                                                  |
| --measure-lagrangian   | M10: Full SM Lagrangian card (enables M1–M9)                         |
| --measure-hausdorff    | M11: Hausdorff dimension (BFS volume growth)                         |
| --measure-zigzag       | M12: Zigzag KK dimension                                             |
| --measure-collider     | M13: Topological collider ( $Q_{\text{topo}}$ )                      |
| --measure-mass-formula | M14: Exact mass formula (genus ladder)                               |
| --measure-all          | Enable all M1–M9, M11–M14                                            |
| --topology-only        | Skip Phase 3 walks; auto-enable M11/M13/M14 ( $\sim 5\times$ faster) |

---

## 16.2.5 Modulo Configuration

| Flag                 | Default | Description                 |
|----------------------|---------|-----------------------------|
| --modulo-prime <u64> | 65537   | Prime modulus $p$ for M3/M6 |
| --modulo-root <u64>  | 3       | Primitive root $g$ of $p$   |
| --modulo-steps <N>   | 500     | NTT walk steps              |

---

## 16.2.6 Export

| Flag                  | Description                                            |
|-----------------------|--------------------------------------------------------|
| --export-slice <PATH> | Export topology slice (single realisation, early exit) |
| --help                | Show usage help                                        |

---

## 16.3 Quick-Start Examples

### 16.3.1 Minimal Run

```
cargo run --release --bin feg_prism -- \
  5000 8 ./output --inmemory --seed 42
```

Runs  $N = 5000$ ,  $M \leq 8$  realisations, deterministic seed, in-memory mode.

### 16.3.2 Full Standard Model Card

```
cargo run --release --bin feg_prism -- \
  50000 25 ./sm_run --inmemory \
  --measure-lagrangian --seed 42
```

Enables all M1–M10 measurements and produces the complete SM parameter card.

### 16.3.3 Production Run

```
cargo run --release --bin feg_prism -- \
  1000000 50 ./production --inmemory \
  --measure-all --target-error 0.005 --seed 42
```

One million events, up to 50 realisations, tighter convergence target.

### 16.3.4 Born Rule Verification

```
cargo run --release --bin feg_prism -- \
  5000 10 ./born --inmemory \
  --measure-decoherence --seed 42
```

### 16.3.5 PMNS Mixing Matrix

```
cargo run --release --bin feg_prism -- \
  50000 25 ./pmns --inmemory \
  --measure-pmns --seed 42
```

Automatically enables M7 (neutrino census) as a prerequisite.

### 16.3.6 Higgs Mass Estimate

```
cargo run --release --bin feg_prism -- \
  50000 25 ./higgs --inmemory \
  --measure-higgs --seed 42
```

### 16.3.7 Animation Export

```
cargo run --release --bin feg_prism -- \
  5000 1 ./anim --inmemory \
  --export-slice ./anim/slice.bin --seed 42
```

Early exit after Phase 2; no spectral walkers run.

### 16.3.8 Resume from Checkpoint

```
cargo run --release --bin feg_prism -- \
  50000 50 ./production --inmemory \
  --measure-all --resume --seed 42
```

Continues from the last checkpoint in *./production/*.

### 16.3.9 Custom Modulo Parameters

```
cargo run --release --bin feg_prism -- \
  5000 10 ./modulo --inmemory \
  --measure-modulo --modulo-prime 257 \
  --modulo-root 3 --modulo-steps 1000 --seed 42
```

# Expected Results & Verification

## Chapter Overview

- Expected output values at standard scales
- Verification checklist
- Known finite-size effects

## 17.1 Spectral Dimension

### Key Result

At  $N = 50,000$ ,  $M = 25$ :

- $d_S(t \rightarrow 0) \rightarrow 2$  (short-distance fractal)
- $d_S(t \rightarrow \infty) \rightarrow 4$  (long-distance manifold recovery)
- The lattice-to-continuum crossover occurs at  $t \sim D_{\max} = 15$  diffusion steps, consistent with the lattice mode decay timescale. Below this,  $d_S(t)$  exhibits lattice artifacts (degree echoes, parity oscillations); above this,  $P(t) \sim t^{-2}$  scaling emerges
- Defect  $d_S$  is slightly reduced compared to vacuum (matter introduces topological shortcuts)

## 17.2 Mass Hierarchy

### Key Result

Cover-time mass ratios ( $M1$ ,  $N = 50,000$ ,  $M = 25$ ):

- $m_\mu/m_e \approx 200 \pm 30$  (target: 206.77, PDG)
  - $m_\tau/m_e \approx 3500 \pm 500$  (target: 3477.4, PDG)
- Exact agreement improves with  $N$ ; finite-size corrections scale as  $N^{-1/4}$ .

## 17.3 Fine Structure Constant

### Key Result

Topological collider (strict bipartite seam + external Markov blanket):

- $Q_{\text{topo}} = 1/4$  (exact, locked asymptote across  $N = 10^4$  to  $10^7$ )
- $\alpha_0 = Q_{\text{topo}}/(8\pi) = 1/(32\pi) \approx 1/100.5$  (bare Planck coupling)
- Screening ratio  $32\pi/137 \approx 0.733$ : vacuum polarization to lab value

## 17.4 Dark Matter Fraction

### Key Result

Phase-coherence decomposition ( $N = 50,000$ ,  $M = 25$ ):

- $\Omega_{\text{dark}}/\Omega_{\text{vis}} \approx 4\text{--}6$  (target: 5.3, Planck)
- Sterile prisms ( $\Phi = 0$ ) constitute  $\sim 10\text{--}20\%$  of all prisms

## 17.5 Born Rule

### Key Result

*Decoherence measurement ( $M6$ ,  $N = 5,000$ ,  $M = 10$ ):*

- *Pearson  $r$  between predicted  $|\psi|^2$  and observed  $K_{2,2}$  PMF validates Born rule*
- *Permutation null model (1000 shuffles) confirms significance*
- *Coherence decay correlation vs  $1/\sqrt{\text{env\_size}}$  measures decoherence rate*

## 17.6 Verification Checklist

1. Compile: `cargo build --release --bin feg_prism`
2. Provenance: startup message shows “SHA-256 verified ✓”
3. Triangle-freeness: `topology_summary.csv` shows zero  $K_4$  count
4. Prism counts:  $\sim 5\text{--}15\%$  of core nodes form prisms
5. Generation hierarchy:  $\text{Gen1} > \text{Gen2} > \text{Gen3}$  in abundance
6. Spectral dimension:  $d_S(1) \approx 2$ ,  $d_S(30) \approx 4$
7. CSV files: all provenance headers present and hash matches
8. Determinism: same seed produces identical output

## 17.7 Known Finite-Size Effects

- **Boundary effects:** The causal diamond has sharp boundaries. Nodes near the tips have anomalously low degree. The core selection (top 10% by degree) mitigates this by preferring interior nodes.
- **Mass ratio convergence:** The cover-time ratios  $m_\mu/m_e$  and  $m_\tau/m_e$  converge slowly (as  $N^{-1/4}$ ). For precision to within 10%, use  $N \geq 50,000$ .
- **Spectral dimension plateau:** The  $d_S \rightarrow 4$  plateau requires  $t \gtrsim D_{\max} = 15$  (lattice mode decay timescale). The default  $t_{\max} = 15$  calibrates the CRT budget at the onset of this plateau; the full step schedule extends to  $t = 100$  and auto-convergence ensures the observable at  $t \approx 36$  stabilises.
- **Ensemble variance:** Some observables (e.g.,  $\alpha_{\text{EM}}$ ) have high variance across realisations. Use  $M \geq 25$  for reliable estimates.

# Dependencies

## Chapter Overview

- *Nine Rust crates*
- *No external SHA-256 dependency*
- *Build profile for performance*

## 18.1 Crate List

| Crate      | Version | Purpose                                            |
|------------|---------|----------------------------------------------------|
| nalgebra   | 0.33    | Dense linear algebra (eigendecomposition)          |
| sprs       | 0.11    | Sparse matrix operations ( $A^2$ for Tier 1 Hasse) |
| rand       | 0.8     | Pseudorandom number generation (StdRng)            |
| rayon      | 1.10    | Data parallelism (parallel iterators)              |
| sysinfo    | 0.33    | System memory detection                            |
| smallvec   | 1.15.1  | Stack-allocated small vectors                      |
| rustc-hash | 2       | Fast non-cryptographic hashing                     |
| serde      | 1       | Serialisation framework (with derive)              |
| bincode    | 1.3     | Binary serialisation for animation export          |

## 18.2 Build Profile

```
[profile.release]
opt-level = 3
lto = "thin"
codegen-units = 1
```

This maximises performance at the cost of longer compile times. The `codegen-units = 1` setting enables whole-program optimisation.

## 18.3 Notable Absences

- **No SHA-256 crate:** the 204-line custom implementation in *provenance.rs* avoids any external cryptographic dependency.
- **No TOML parser:** the configuration reader in *config.rs* uses simple line-by-line parsing (12 lines).
- **No logging framework:** diagnostics use `println!` with optional file-append to *simulation.log*.



# Colophon

This document was typeset with XeLaTeX using the Modulo Synthesis palette and tcolor-box theorem environments, matching the styling of the pedagogic booklet series *Modulo Synthesis, Volumes I & II*.

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